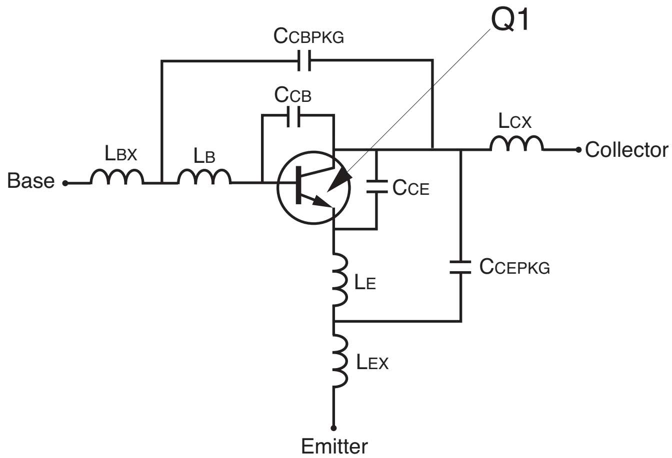
CEL

NONLINEAR MODEL | NE68019

SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS (1)

Parameters	Q1	Parameters	Q1
IS	3.84e-16	MJC	0.64
BF	124.9	XCJC	0
NF	1.05	CJS	0
VAF	11.9	VJS	0.75
IKF	0.027	MJS	0
ISE	1.0e-14	FC	0.5
NE	2.17	TF	8.7e-12
BR	1	XTF	18
NR	1.05	VTF	19.1
VAR	Infinity	ITF	0.082
IKR	Infinity	PTF	0
ISC	0	TR	0.635e-9
NC	2	EG	1.11
RE	0.6	XTB	0
RB	17.9	XTI	3
RBM	1.02	KF	0
IRB	4.01e-4	AF	1
RC	10.5		
CJE	0.358e-12		
VJE	0.71		
MJE	0.5		
CJC	0.162e-12		
VJC	0.79		

(1) Gummel-Poon Model

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

ADDITIONAL PARAMETERS

Parameters	68019
C_{CB}	0.08e-12
C_{CE}	0.08e-12
L_B	0.72e-9
L_E	0.76e-9
C_{CBPKG}	0.17e-12
C_{CEPKG}	0.21e-12
L_{BX}	0.19e-9
L_{CX}	0.19e-9
L_{EX}	0.19e-9

MODEL RANGE

Frequency: 0.05 to 3.0 GHz

Bias: V_{CE} = 1 V to 6 V, I_C = 1 mA to 15 mA

Date: 8/03

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